

SIMATS ENGINEERING

CO 3 - ASSESSMENT TOOL 2

Data Interpretation

NAME : B. TARUNREDDY

REG.NO : 192424213

COURSE NAME: FUNDAMENTALS OF DATA SCIENCE

COURSE CODE: DSA0404

FACULTY NAME : DR.KRISHNARAJ MONY

6. Sampling Distribution Reasoning

Question:

A population has mean $\mu = 500$ and SD $\sigma = 80$. A random sample of $n = 64$ is drawn.

- (a) Compute the standard error of the sample mean.
- (b) What is the probability that the sample mean exceeds 520, assuming the sampling distribution is approximately normal?
- (c) If n were increased to 256, how does the standard error change, and what principle explains this?

Solution:

(a) Standard Error of the Sample Mean

Given: $\mu = 500$, $\sigma = 80$, $n = 64$

Formula: $SE(\bar{X}) = \sigma / \sqrt{n}$

$$SE(\bar{X}) = 80 / \sqrt{64}$$

$$SE(\bar{X}) = 80 / 8$$

$$SE(\bar{X}) = 10$$

Answer: $SE(\bar{X}) = 10$

(b) Probability that the Sample Mean Exceeds 520

We need: $P(\bar{X} > 520)$

First calculate the Z-score:

$$Z = (\bar{X} - \mu) / SE$$

$$Z = (520 - 500) / 10$$

$$Z = 20 / 10 = 2$$

From the standard normal table: $P(Z < 2) = 0.9772$

Therefore: $P(Z > 2) = 1 - 0.9772 = 0.0228$

Answer: $P(\bar{X} > 520) = 0.0228 = 2.28\%$

(c) Effect of Increasing the Sample Size

If $n = 256$:

$$SE = \sigma / \sqrt{n}$$

$$SE = 80 / \sqrt{256}$$

$$SE = 80 / 16$$

$$SE = 5$$

Answer: The standard error decreases from 10 to 5.

Principle: $SE \propto 1/\sqrt{n}$. As sample size increases, the sample mean becomes more precise. This is related to the Law of Large Numbers and the behavior of sampling distributions.

7. Covariance Calculation and Sign Interpretation

Question:

Given paired data on advertising spend (X, ₹'000) and sales (Y, ₹ lakh):

X (₹'000)	Y (₹ lakh)
10	15
20	18
30	25
40	30
50	42

Calculate $\text{Cov}(X,Y)$. Is the relationship direction consistent with intuition? What does the magnitude alone fail to tell you (that correlation would)?

Solution:

Step 1: Calculate the mean of X

$$\bar{X} = (10 + 20 + 30 + 40 + 50) / 5$$

$$\bar{X} = 150 / 5 = 30$$

Step 2: Calculate the mean of Y

$$\bar{Y} = (15 + 18 + 25 + 30 + 42) / 5$$

$$\bar{Y} = 130 / 5 = 26$$

Step 3: Calculate deviations and products

X	Y	X - 30	Y - 26	(X-30)(Y-26)
10	15	-20	-11	220
20	18	-10	-8	80
30	25	0	-1	0
40	30	10	4	40
50	42	20	16	320
Total				660

Step 4: Calculate population covariance

$$\text{Cov}(X,Y) = \Sigma(X - \bar{X})(Y - \bar{Y}) / n$$

$$\text{Cov}(X,Y) = 660 / 5$$

$$\text{Cov}(X,Y) = 132$$

Answer: $\text{Cov}(X,Y) = +132$

Interpretation: The positive covariance indicates that advertising spend and sales move in the same direction. This is consistent with intuition: higher advertising spend is associated with higher sales.

What covariance magnitude alone fails to tell us: Covariance depends on the units and scale of the variables, so its magnitude does not provide a standardized measure of relationship strength. Correlation gives a standardized value between -1 and $+1$ and therefore makes strength easier to interpret and compare.

8. Correlation vs Causation Trap

Question:

A city reports that ice cream sales and drowning incidents have a correlation coefficient of $r = 0.85$. A newspaper claims "ice cream causes drowning." Identify the statistical flaw, name the likely confounding variable, and explain what additional evidence (not just correlation) would be needed to claim causation.

Solution:**Step 1: Interpret the correlation**

$r = 0.85$ indicates a strong positive correlation between ice cream sales and drowning incidents.

Step 2: Identify the statistical flaw

The newspaper commits the error: correlation does not imply causation. A correlation only shows that two variables move together; it does not prove that one variable causes the other.

Step 3: Identify the likely confounding variable

Likely confounding variable: temperature or hot weather.

During hot weather, ice cream sales increase and more people may go swimming or spend time near water, increasing exposure to drowning risk.

Relationship:

Hot weather $\rightarrow$ higher ice cream sales

Hot weather $\rightarrow$ more swimming/water exposure $\rightarrow$ potentially more drowning incidents

Step 4: Evidence needed for causation

Correlation alone is insufficient. Stronger evidence would include controlled or randomized experiments where ethical and feasible, longitudinal evidence, careful control of confounding variables, consistent findings across multiple studies, and a plausible causal mechanism.

Answer: The claim is a correlation-versus-causation error. The likely confounder is hot weather/temperature. Additional causal evidence is required.

9. Effect of Outliers on Correlation

Question:

A dataset of 9 points shows $r = 0.42$ (weak positive). One extreme outlier is removed, and r becomes 0.89. What does this reveal about the fragility of Pearson's correlation? Would Spearman's rank correlation be more robust here? Justify.

Solution:**Step 1: Compare the correlations**

Before removing the outlier: $r = 0.42$

After removing the outlier: $r = 0.89$

Step 2: Interpret the change

The correlation changes substantially from 0.42 to 0.89 after removing only one point.

This shows that Pearson's correlation can be highly sensitive to extreme or influential observations.

Answer: Pearson's correlation is sensitive to outliers, and a single influential point can substantially change the estimated linear relationship.

Step 3: Spearman's rank correlation

Yes, Spearman's rank correlation would generally be more robust to extreme numerical values because it is calculated using ranks rather than the raw magnitudes of the observations.

However, Spearman is not completely immune to outliers because an unusual observation can still affect its rank.

Final Answer: The large change from 0.42 to 0.89 demonstrates the fragility of Pearson's correlation to influential outliers. Spearman's rank correlation is generally more robust in this situation.

10. Data Preparation — Missing Values

Question:

A dataset of 200 rows has 15% missing values in the "Income" column, and analysis shows these missing values are more common among unemployed respondents (not random).

- (a) Classify this missing data mechanism (MCAR/MAR/MNAR).
- (b) Why would mean imputation introduce bias here?
- (c) Suggest a better strategy.

Solution:

Step 1: Calculate the number of missing values

Missing values = $200 \times 15\%$

Missing values = 200×0.15

Missing values = 30

Therefore, 30 Income values are missing.

(a) Classify the missing-data mechanism

Answer: MAR — Missing At Random.

Reason: The probability that Income is missing is related to an observed variable, employment status. Therefore, the missingness is not completely random.

(b) Why mean imputation can introduce bias

The missing Income values are more common among unemployed respondents. Unemployed respondents may have income characteristics that differ from the overall population.

If the overall mean income is used for all missing values, it ignores the relationship between employment status and income.

For example, if unemployed respondents generally have lower income, replacing their missing values with the overall mean would likely overestimate their income.

Mean imputation can also distort the income distribution and underestimate variability.

(c) Better strategy

A better approach is to use a method that accounts for employment status and other relevant observed variables.

Option 1: Group-based imputation — calculate separate income estimates for employed and unemployed groups.

Option 2: Multiple imputation — estimate missing income using predictors such as employment status, age, education, occupation, and other relevant variables.

Final Answer: (a) MAR. (b) Mean imputation can introduce bias because missingness is related to employment status. (c) Use group-based imputation or, preferably, multiple imputation using relevant predictors.